

Stride Intelligence

IMA-derived movement-quality signals — coach guide

What it does

Stride Intelligence reads four signals from the Catapult Vector S7+ vest and turns them into early-warning flags about a player's movement quality. It works the same way indoor and outdoor — wherever the vest captures stride and IMU events.

The four signals:

- Cadence — how fast the player's legs are turning over
- Stride length at HSR — how much ground each sprint stride covers
- Left/right cutting asymmetry — is the player favoring one side
- GPS-IMA decoupling — is the player covering distance with less mechanical effort than usual

Each signal compares today's value against the player's own 28-day baseline (or, for asymmetry, against published thresholds). If something drifts, the player gets a flag in the Decision Summary.

Why it matters

1. Catches NM fatigue before the player tells you

Cadence drops 1-2 days before a player verbally complains of heavy legs. Stride length compression often shows 2-5 days before hamstring/quad strain. By the time soreness rises in self-report, the mechanical compensation has already been there for days.

2. RTP clearance gate

Bishop 2020 showed L/R asymmetry > 15% on cutting tasks more than doubles ACL re-injury risk on return to play. The accepted clearance target is < 10% asymmetry sustained for 5-7 days. Stride Intelligence gives you that number per session.

3. Validates GPS readings

Two players running 22 km/h can have very different mechanical costs. If GPS shows high speed but IMU stride load stays low, the player is coasting or compensating. If GPS speed and IMU effort line up, you're seeing genuine effort.

The four metrics

1. Cadence (Hz)

Volume-weighted average stride frequency across all 8 IMA Free Running bands.

Formula: $\Sigma(\text{stride_count} \times \text{stride_rate}) / \Sigma \text{stride_count}$

- **Normal range:** 2.5-3.2 Hz at running speed for adult footballers
- **Flag triggers:** today's cadence is more than 1σ below the player's 28-day mean
- **What it means:** legs are turning over slower than usual — early NM fatigue indicator (Cardinale 2017)
- **Action:** monitor closely, consider reducing high-intensity intervals next session

2. Stride length at high-speed running (m/stride)

How many meters of HSR distance the player covers per high-velocity stride.

Formula: $\text{HSR_distance} / \text{count}(\text{strides_in_band_5_to_8})$

- **Normal range:** 1.8-2.2 m/stride at sprint speed (varies by player height)
- **Flag triggers:** today's value is more than 1σ below the player's baseline
- **What it means:** strides are shorter than usual — compensation pattern, often 2-5 days before hamstring or quad strain
- **Action:** flag for hamstring/groin assessment, reduce sprint volume, consider medical follow-up if pattern persists 3+ days

3. Left/right cutting asymmetry (%)

Difference between left-side and right-side change-of-direction events, normalized.

Formula: $|L - R| / ((L + R) / 2) \times 100$

- **Source data:** ima_band1+2+3 left/right counts (always populated by Catapult IMU)
- **Minimum events:** ≥ 5 total CoD events for the % to be reliable
- **Bishop 2020 thresholds:** < 9% balanced · 9-15% watch · 15-18% concern · > 18% high
- **RTP clearance:** < 10% sustained for 5-7 days before returning to full training
- **Action:** > 15% — flag for medical/S&C review, investigate before next match

4. GPS-IMA decoupling (percentage points)

Difference between the share of distance covered at high speed (GPS) and the share of player load coming from high-intensity strides (IMU).

Formula: $(\text{HSR_distance} / \text{total_distance}) - (\text{high_band_player_load} / \text{total_player_load})$

- **Normal range:** close to zero, varies by player baseline
- **Flag triggers:** today's value is more than 1σ above the player's baseline (positive direction)
- **What it means:** the player is covering distance with less mechanical cost than usual — coasting on momentum, or NM fatigue suppressing effort
- **Action:** compare against subjective wellness; if low energy / soreness aligns, treat as cumulative fatigue signal

Severity levels

Level	Personal-z	Asymmetry	Action
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Watch	-1.0σ to -1.5σ	9% to 15%	Note in coach summary, monitor next session
Concern	-1.5σ to -2.0σ	15% to 18%	Reduce high-intensity exposure, flag for S&C
High	$\leq -2.0\sigma$	$\geq 18\%$	Investigate before next match, consider medical follow-up

Where to find it in MicroPulse

Stride Intelligence appears in two places:

Decel Intelligence page

Click any player row to expand. The Stride Intelligence card sits below the AI Decel Summary. It shows:

- Driver chips (watch / concern / high) with σ -score or % value
- Coach-readable reason text per fired flag
- Four metric tiles with 14-day sparklines and personal baselines

Decision Summary

When any stride flag fires, the verdict text on the main coach dashboard includes the reason. The drivers list also shows entries like 'Cadence below personal baseline', 'Stride length compressing', 'Left/right cutting asymmetry', or 'GPS-IMA decoupling (effort vs distance)'.

How baselines are built

Each metric uses a 28-day rolling personal baseline computed nightly at 02:05. Until the player has at least 14 days of data, the baseline status is 'calibrating' and flags don't fire (status 'active' kicks in at $n \geq 14$).

- ≤ 5 days: insufficient data, no flags
- 6-13 days: calibrating, no flags but data is being collected
- ≥ 14 days: active, drift detection runs every day

L/R asymmetry uses absolute thresholds (Bishop 2020), so it works as soon as the player has ≥ 5 CoD events in a single session.

Coach actions by driver

Driver	What you do
Cadence drop	Note in coach summary. If sustained 2+ days, reduce HIIT/sprint exposure next session by 20-30%.
Stride compression	Flag for hamstring / quad / groin assessment. Avoid maximal sprints. Re-check tomorrow.
L/R asymmetry $> 15\%$	Investigate before next match. Could be a pre-existing imbalance, fresh injury, or compensation.

	Cross-check with VALD ForceFrame if available.
GPS-IMA decoupling	Cross-check with subjective wellness. If energy and soreness scores have dropped too, treat as cumulative fatigue. Reduce session load.

Frequently asked

Why does cadence sometimes show 0?

The IMA Free Running pipeline only generates data when the player runs for sustained periods. Recovery days, gym sessions, and warmup-only days will show no stride data — and the card shows 'no data' rather than fake numbers.

Can stride length be too high?

Yes, but it's rarer. Overstriding shows as stride length above baseline + cadence below baseline = each step is bigger but slower. This is also a fatigue compensation pattern. The card flags both directions, severity weighted toward the dominant driver.

What if a player has only one match per week?

Baselines need 14 days minimum, so a player who only trains once a week will take 14 weeks before the active flags kick in. Use the raw values on the card for comparison even when status is 'calibrating' — you can still spot relative changes from one week to the next.

Does this work indoor (futsal/handball)?

Yes — IMA Free Running is IMU-only and doesn't need GPS. The cadence, stride length proxy, and L/R asymmetry all populate identically indoor. GPS-IMA decoupling needs at least some GPS distance to compute, so it's outdoor-only.

References

- Bishop C et al. (2020). Asymmetry thresholds for screening tests in professional soccer.
- Cardinale M et al. (2017). Cadence as neuromuscular fatigue indicator in team sport.
- McBurnie A et al. (2022). Deceleration biomechanics and ACL/quadriceps injury risk.
- Catapult IMA Free Running White Paper — stride-band detection methodology.